**DATASCI 406 FA 26, Quiz 2.1**

| Name | Uniqname | UMID |
|---|---|---|
| | | |

Each question is worth 1 point. Circle one answer for each.

## Question 1 (1 point)

For a random variable $X$ and a probability $p$, what does the quantile function $Q_X(p)$ return?

1. The probability that $X$ is less than or equal to $p$.
2. The smallest value $x$ such that the CDF evaluated at $x$ is at least $p$.
3. The value $x$ at which the density of $X$ equals $p$.
4. The value of the CDF evaluated at $X$.

## Question 2 (1 point)

$X$ has CDF $F(x) = x^3$ for $0 \le x \le 1$. What is its quantile function $Q(u)$ for $0 \le u \le 1$?

1. $u^3$ 2. $3u^2$ 3. $1 - u^{1/3}$ 4. $u^{1/3}$

## Question 3 (1 point)

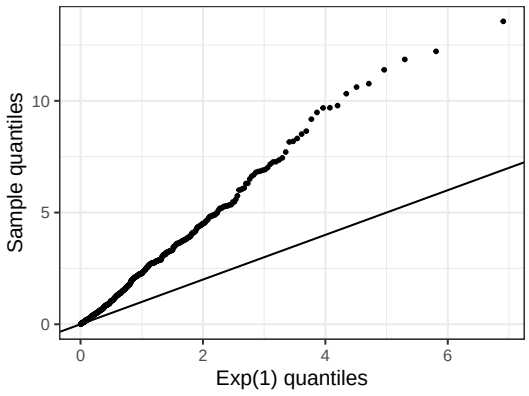
Which line of R produces 1000 draws from an exponential distribution with rate 2, using `qexp()` and not `rexp()`?

1. `runif(1000, qexp(0, rate = 2), qexp(1, rate = 2))`
2. `qexp(1000, rate = 2) * runif(1000)`
3. `qexp(runif(1000), rate = 2)`
4. `runif(qexp(1000, rate = 2))`

## Question 4 (1 point)

A sampler is meant to draw from Exp(1). The QQ plot at right compares 500 of its draws with the Exp(1) quantiles, and the line is the 45 degree line. Is the sampler correct?

1. Correct: the points form a straight line.
2. Wrong: the points do not follow the 45 degree line.
3. Can't tell: a plot is not a hypothesis test.




## Question 5 (1 point)

$X$ takes the values below. To draw $X$ you generate $u = 0.53$ and apply the inverse transform method for discrete variables. What value $x$ do you generate?

| $x$ | 10 | 20 | 30 | 40 |
|---|---|---|---|---|
| $\Pr(X = x)$ | 0.2 | 0.3 | 0.1 | 0.4 |

1. 10 2. 20 3. 30 4. 40